



**ROSEBANK
INTERNATIONAL**
University College

RaceDay

Part 1 – Database Design and API Planning

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1. RaceDay Requirements

RaceDay digitises event management for road running, walking, and cycling events in South Africa, replacing paper based registration and disconnected spreadsheets. The system supports exactly two user roles:

Event Organiser

- Create, edit, and delete events.
- Manage event categories.
- Capture participant results.
- View event enrolments.

Participant

- Create an account.
- Browse upcoming events.
- Enter an event by selecting a category.
- View their own enrolments.
- Check their own personal results.

2. Business Rules

Account and Role Rules

1. A Participant must be able to create an account before entering an event.
2. RaceDay must support two main roles: Event Organiser and Participant.
3. An Event Organiser may create events.
4. An Event Organiser may manage the events they are responsible for.
5. A Participant may browse available events.
6. A Participant may view their own enrolments and personal results.
7. Participants must not be allowed to perform Event Organiser functions such as creating or managing events and categories.

Event Rules

8. An event is managed by an Event Organiser.

- 9.** An Event Organiser must be able to create an event.
- 10.** An Event Organiser must be able to edit an event they manage.
- 11.** An Event Organiser must be able to delete an event they manage.
- 12.** An event may have one or more categories that Participants can select when entering the event.
- 13.** Participants must be able to browse available events.

Category Rules

- 14.** An Event Organiser must be able to create categories for an event.
- 15.** An Event Organiser must be able to manage categories belonging to an event they manage.
- 16.** A category must belong to an event.
- 17.** A Participant enters an event by selecting an available category for that event.

Enrolment Rules

- 18.** An enrolment records a Participant entering an event.
- 19.** Each enrolment must be linked to a Participant.
- 20.** Each enrolment must be linked to an event.
- 21.** Each enrolment must record the category selected by the Participant.
- 22.** A Participant must be able to view their own event enrolments.
- 23.** An Event Organiser must be able to view the enrolments for events they manage.
- 24.** Participants must not be able to view or manage other Participants' enrolments.

Result Rules

- 25.** An Event Organiser must be able to capture participant results.
- 26.** A participant result must be associated with the appropriate Participant and event enrolment.
- 27.** An Event Organiser must be able to manage results for participants in events they manage.
- 28.** A Participant must be able to check their own results.
- 29.** Participants must not be able to view or manage another Participant's personal results.

Role-Based Access Rules

- 30.** RaceDay must restrict Event Organiser functions to authorised Event Organiser accounts.
- 31.** RaceDay must restrict Participant functions to authorised Participant accounts.
- 32.** Participants may enter events, view their enrolments and check their own results.
- 33.** Event Organisers may create and manage events, manage event categories, capture participant results and view event enrolments.

Business Rule Summary

The RaceDay system uses the Event Organiser and Participant roles to control access to

system functionality. Event Organisers are responsible for managing events, categories, enrolments and participant results, while Participants can create accounts, browse events, enter

Section A - Entity Relationship Diagram (ERD)

The RaceDay database consists of six entities, derived directly from the Business Rules above: Organisers, Participants, Events, Categories, Enrolments, and Results. Each entity, its attributes, keys, and relationships are detailed below.

Organisers

Represents the required Admin table. The POE Background's Event Organiser role is preserved as the functional role name; "Admin" is the naming convention requested for the submission structure only.

Attribute	Data Type	PK	FK / Constraint	Explanation
OrganiserId	INT	PK	-	Unique identifier for role-based access.
FullName	NVARCHAR(100)			Identifies the account holder.
Email	NVARCHAR(150)		UNIQUE	Used for login/authentication.
PasswordHash	NVARCHAR(255)			Used for authentication.
PhoneNumber	VARCHAR(20)			
CreatedAt	DATETIME		DEFAULT	

Participants

Represents the required User table. The POE Background's Participant role is preserved as the functional role name; "User" is the naming convention requested for the submission structure only.

Attribute	Data Type	PK	FK / Constraint	Explanation
ParticipantId	INT	PK	-	Unique identifier for role-based access.
FullName	NVARCHAR(100)			Identifies the account holder.
Email	NVARCHAR(150)		UNIQUE	Used for login/authentication.
PasswordHash	NVARCHAR(255)			Used for authentication.
PhoneNumber	VARCHAR(20)			
CreatedAt	DATETIME		DEFAULT	

Events

Attribute	Data Type	PK	FK / Constraint	Explanation
EventId	INT	PK	-	Unique identifier for the event.
OrganiserId	INT		FK -> Organisers.OrganiserId	An event is managed by exactly one Organiser.

Attribute	Data Type	PK	FK / Constraint	Explanation
EventName	NVARCHAR(150)			Needed for Participants to browse events.
EventDate	DATE			A bookable event needs a date.
Location	NVARCHAR(150)			Supports browsing
Status	VARCHAR(20)		DEFAULT	

Categories

Carries a composite UNIQUE constraint on (EventId, CategoryId) so that Enrolments can enforce that a selected category belongs to the same event recorded on the enrolment.

Attribute	Data Type	PK	FK / Constraint	Explanation
CategoryId	INT	PK	UQ (EventId, CategoryId)	Unique identifier for the category.
EventId	INT		FK -> Events.EventId	A category must belong to an event.
CategoryName	NVARCHAR(100)			Participant selects this when entering an event.
DistanceKm	DECIMAL(5,2)			-
EntryFee	DECIMAL(8,2)		DEFAULT	-

Enrolments

Associative entity resolving the Participant-Event-Category relationship. The composite foreign key on (EventId, CategoryId) referencing Categories(EventId, CategoryId) makes it structurally impossible for an enrolment to reference a category from a different event.

Attribute	Data Type	PK	FK / Constraint	Explanation
EnrolmentId	INT	PK	-	Unique identifier for the enrolment.
ParticipantId	INT		FK -> Participants.ParticipantId	Each enrolment belongs to one Participant.
EventId	INT		FK (composite)	Each enrolment is for one Event.
CategoryId	INT		FK (composite) -> Categories(EventId, CategoryId)	Category is validated against the same event.
EnrolmentDate	DATETIME		DEFAULT	-
Status	VARCHAR(20)		DEFAULT	-

Results

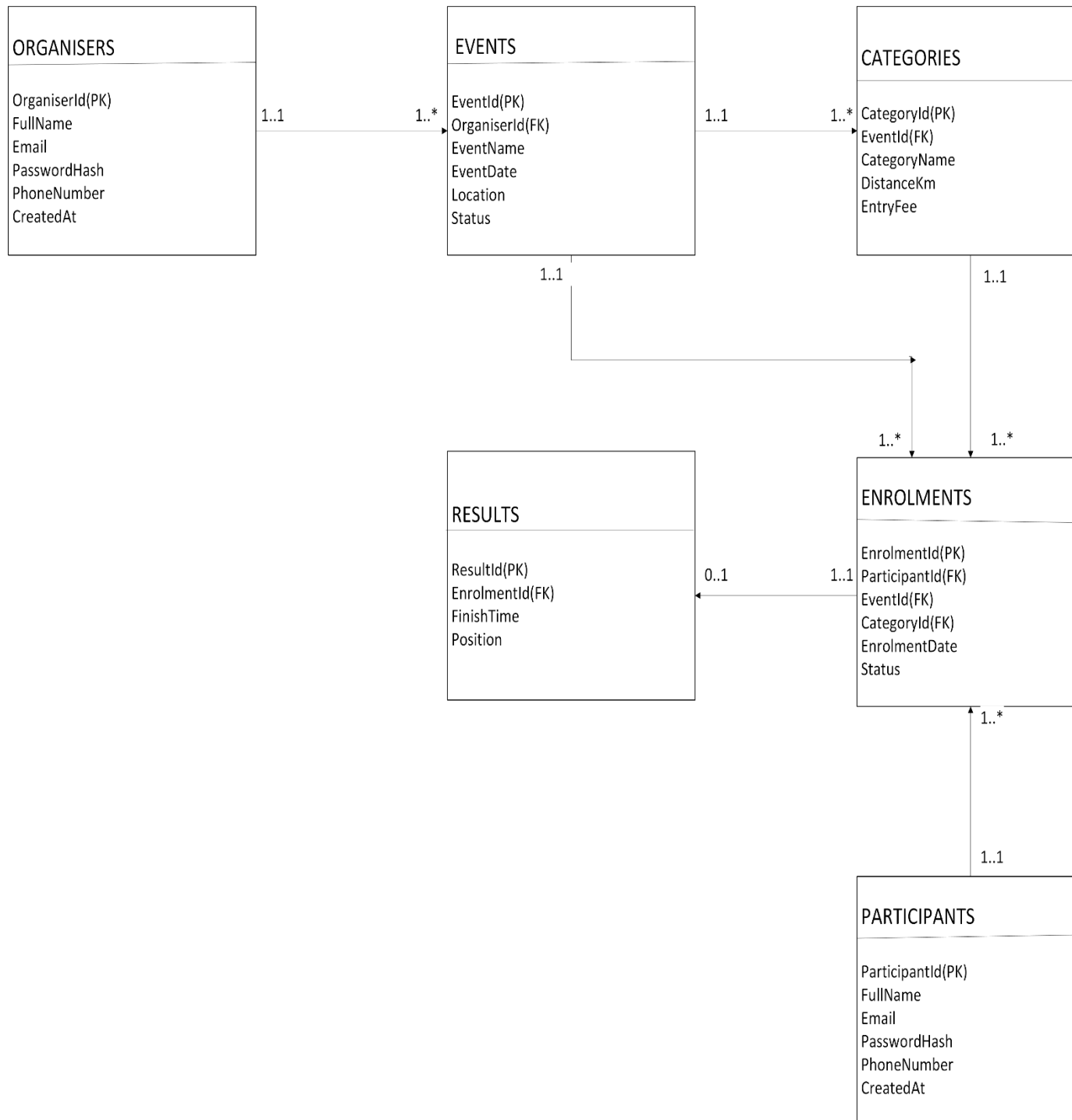
EnrolmentId is NOT NULL and UNIQUE. This enforces that each Result belongs to exactly one Enrolment and that an Enrolment can have at most one Result. An Enrolment may exist without a Result.

Attribute	Data Type	PK	FK / Constraint	Explanation
ResultId	INT	PK	-	Unique identifier for the result.
EnrolmentId	INT		FK -> Enrolments.EnrolmentId, UNIQUE	Links the result to exactly one enrolment.
FinishTime	TIME			[Assumption] Natural minimal field for a race result.
Position	INT			[Assumption] Natural minimal field for a race result.

Relationships and Cardinalities

Entity A	Entity B	Cardinality	Explanation
Organisers	Events	1 -> M	One Organiser manages many Events; each Event has exactly one Organiser.
Events	Categories	1 -> M	One Event has many Categories; each Category belongs to exactly one Event.
Participants	Enrolments	1 -> M	One Participant makes many Enrolments; each Enrolment belongs to one Participant.
Events	Enrolments	1 -> M	One Event receives many Enrolments; each Enrolment is for one Event.
Categories	Enrolments	1 -> M	One Category is selected in many Enrolments, scoped to the same Event.
Enrolments	Results	1 -> 0..1	An Enrolment may or may not yet have a Result.

ERD Diagram



Section B - API Endpoint Plan

The API endpoint plan below covers all six required functional areas: Authentication, User Profile, Events, Categories, Event Enrolments, and Results. It contains 33 endpoints in total, each mapped directly to a business rule. Ownership restrictions are noted in the Role Required column where the user can access only their own data or events they manage.

Authentication (Register & Login)

Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register/participant	Creates a new Participant account.	Public	FullName, Email, Password	201 Created, new ParticipantId
POST	/api/auth/register/organiser	Creates a new Organiser account.	Public	FullName, Email, Password	201 Created, new OrganiserId
POST	/api/auth/login/participant	Logs in a Participant and returns an access token.	Public	Email, Password	200 OK, auth token
POST	/api/auth/login/organiser	Logs in an Organiser and returns an access token.	Public	Email, Password	200 OK, auth token
POST	/api/auth/refresh-token	Issues a new access token using a valid refresh token.	Participant, Organiser	RefreshToken	200 OK, new token
POST	/api/auth/logout	Logs out the currently authenticated user.	Participant, Organiser	None	200 OK

User Profile

Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/participants/{id}	Retrieves the logged-in Participant's own profile.	Participant	None	200 OK, profile data
PUT	/api/participants/{id}	Updates the logged-in Participant's own profile.	Participant	FullName, PhoneNumber	200 OK, updated profile

GET	/api/organisers/{id}	Retrieves the logged-in Organiser's own profile.	Organiser	None	200 OK, profile data
PUT	/api/organisers/{id}	Updates the logged-in Organiser's own profile.	Organiser	FullName, PhoneNumber	200 OK, updated profile

Events

Method	Route	Description	Role Required	Request Body
GET	/api/events	Retrieves all events for Participants to browse.	Participant	None
GET	/api/events/{id}	Retrieves the details of a single event.	Participant	None
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	EventName, Location
PUT	/api/events/{id}	Updates an event managed by the logged-in Organiser.	Organiser	EventName, Location
DELETE	/api/events/{id}	Deletes an event managed by the logged-in Organiser.	Organiser	None
GET	/api/organisers/{id}/events	Retrieves all events managed by a specific Organiser.	Organiser	None